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An Adaptive Pricing Framework for Real-Time AI Model Service Exchange

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Artificial intelligence (AI) model services offer remarkable efficiency and automation, engaging customers across various tasks. However, not all AI consumers possess sufficient data to drive AI model training or the specialized knowledge to construct high-performance AI model structures; this has led to a trend in AI model service transactions, a novel facet of the digital economy. Unlike conventional digital products, AI models undergo performance degradation over time. This phenomenon occurs as the training data becomes outdated, leading to a “distribution shift” away from the target distribution of the most recent downstream tasks. This degradation decreases consumer demand, making the AI model less competitive and lowering provider revenue. In this work, we analyze the impact of performance degradation on consumers' demand for AI model services and propose an adaptive pricing framework for service providers to maximize revenue in real-time AI model service exchange. Specifically, We propose an optimal transport (OT) distance-based approach to estimate model performance degradation effectively. Building on this methodology, we implement several practical solutions for predicting changes in future demand rates resulting from current pricing configurations. We then propose a demand-driven AI model update mechanism for service providers to maintain high product demand rates while reducing retraining AI models' costs. We finally propose a reinforcement learning-based pricing mechanism that facilitates adaptive and rapid pricing responses to achieve revenue maximization. Extensive

experiments in both 2-competitor and multi-competitor markets validate our framework, showing a significant revenue advantage over baseline pricing strategies in AI model service transactions.

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Contents

I. Introduction

Data-driven artificial intelligence (AI) models are increasingly utilized in a variety of complex tasks, spanning from industrial operations to everyday human activities, such as weather forecasting, healthcare, finance, and recommendation systems. However, not all users possess sufficient domain knowledge or computational resources to train high-performance AI models that meet their expectations. This resource gap has led to the rise of pre-trained AI models, also known as foundation models, available for purchase on model service exchange platforms like StreamML [1] and GravityAI [2]. These platforms provide an avenue for buyers to acquire models and either deploy them directly or fine-tune them for specific downstream tasks.

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